

FINITE ELEMENT PROCEDURES FOR FLUID–STRUCTURE INTERACTIONS AND APPLICATION TO LIQUID STORAGE TANKS

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Most of the failures of large tanks after severe earthquakes are suspected to have resulted from the dynamic buckling caused by overturning moments of seismically induced liquid inertia and surface slosh waves. In this paper, a nonlinear finite element method is presented which can treat the structural behavior of the tanks in conjunction with fluid, including the dynamics and buckling. Both the formulation and computer implementation aspects are presented. The areas upon which attention is focused are: the mixed Lagrangian–Eulerian kinematical description for modeling fluid subdomains in fluid–structure interaction problems, the finite rotation effects in numerical integration of rate constitutive equations arising in large-deformation analysis and the implicit–explicit finite element techniques for transient analysis. All these nonlinear methodologies have been integrated into a finite element computer code [47]. A number of physically important buckling and dynamic analyses of tanks are being investigated. Various other fluid structure phenomena such as the stability of off-shore structures can be analyzed with this computer program. The proposed nonlinear finite-element procedures can serve as a basis for future developments of various other fluid–structure phenomena such as the transient motion of submerged or partially submerged structures.

1. Introduction

Over the last two decades, significant attention has been devoted to the development of computational methods for fluid–structure interaction. Although considerable progress has been made in recent years in the development and understanding of finite element methods in structural mechanics, in the fluid mechanic area, finite element research is just coming of age. Although much effort has been exerted in recent years, it is fair to say that transient finite element methods for the Navier–Stokes equations have not achieved the speed and versatility of existing finite difference methods.

The ability to solve general classes of fluid–structure interaction problems depends upon the ability to solve the corresponding uncoupled structural and fluid problems, and also the ability to ‘interface’ fluid and structural subdomains.

Finite element methods applied to fluid–structure interaction problems have been made by several investigators. One of the related current research areas is seismic analysis and design procedures for ground support, cylindrical liquid-filled tanks. Presently used methods are mostly based on linear small motion response and do not take satisfactory account of nonlinear effects

observed in the field under the action of strong earthquakes, such as lift-off of tanks from foundations, finite-amplitude sloshing, and nonlinear and inelastic tank response.

This paper deals with the development of finite element procedures and the computer implementation aspects in fluid–structure interactions with particular reference to dynamic and buckling analyses of fluid-filled tanks. Both geometrical and material nonlinearities are accounted for.

An outline of the remainder of the paper follows. In section 2, the mixed Lagrangian–Eulerian description of motion is reviewed. In section 3, the variational equation of the mixed Lagrangian–Eulerian formulation is discussed. In section 4, the linearized equations of motion using Lagrangian description and the numerical integration of a class of rate-type constitutive equation will be discussed. In section 5, the finite element and matrix equations are derived. In section 6, ‘upwind’ finite element obtained by modifying the standard Gauss–Legendre rules are reviewed and a similar upwind scheme is developed for the mixed Lagrangian–Eulerian description. In section 7, shell element and contact/sliding element are discussed. In section 8, the implicit–explicit temporal time integration algorithms

are employed for the time integration of the variation equations. The computer implementation aspects of these matrix equations are also discussed. The application to fluid-structure interaction is presented in section 9. Conclusions and recommendations for further research are presented in section 10.

2. Review of mixed Lagrangian-Eulerian description of motion

The two classical descriptions of motion, Lagrangian and Eulerian, are not adequate for many free-surface flows and problems of fluid-solid-structure interaction involving finite deformation. The Lagrangian description is desirable when interfaces and free-boundaries are important. However it is generally not suitable when the fluid undergoes large displacements since the mesh will become highly distorted. On the other hand, the Eulerian description is desirable when the fluid experiences large distortions, however, it is not compatible with the large displacement of the structure for which the geometries of the boundary are altered.

To circumvent this difficulty, a more versatile description of motion, the mixed Lagrangian-Eulerian description, is introduced. It is based on the previous developments by Amsden and Hirt [1], Donea et al. [10–12], Belytschko and Kennedy [7], Hughes et al. [31], and references therein.

A general theory of mixed Lagrangian-Eulerian description has been introduced in [31]. The same notations will be used in this paper, and the reader is referred to [31] for further details. To simplify the presentation, a summary of partial differential operators and the definition of convective velocity are given as follows [31]:

$$\begin{aligned} \cdot_i &= \partial/\partial x_i, & \cdot' &= (\partial/\partial t)_{x_i} \text{ fixed} \\ & & & \text{('referential derivatives')}, \\ \cdot_i &= \partial/\partial y_i, & \cdot_t &= (\partial/\partial t)_{y_i} \text{ fixed} \\ & & & \text{('spatial derivatives')}, \\ \cdot_i &= \partial/\partial z_i, & \cdot &= (\partial/\partial t)_{z_i} \text{ fixed} \\ & & & \text{('material derivatives')}, \end{aligned} \quad (2.1)$$

and

$$c_j = \dot{u}_j - \dot{u}'_j \quad (2.2)$$

is called the convective velocity.

3. Variational equations of the mixed Lagrangian-Eulerian formulation

A mixed description will be used for large motion fluid flow problems. It is restricted to incompressible viscous fluid. In order to eliminate the unknown pressure and the continuity equation, a 'slightly compressible formulation' [46] is employed in which the continuity equation is dropped and the constitutive equation for the slightly compressible viscous fluid is:

$$\begin{aligned} \tau_{ij} &= - \left(p_0 - \int_0^t \beta \rho \dot{u}_{k,k} dt \right) \delta_{ij} \\ &+ \mu [(\dot{u}_i)_{,j} + (\dot{u}_j)_{,i}], \end{aligned} \quad (3.1)$$

where p_0 is the pressure at $t=0$ and $\beta = \partial p / \partial \rho = \text{constant}$ (see [46] for a discussion).

The variational equation of the mixed Lagrangian-Eulerian formulation is:

$$\begin{aligned} \int_{R_v} (\rho \ddot{u}_i \bar{u}_i + \tau_{ij} \bar{u}_{i,j}) dR_v &= \\ = \int_{R_v} b_i \bar{u}_i dR_v &+ \int_{\partial R_v^h} h_i \bar{u}_i dA_v, \end{aligned} \quad (3.2)$$

in which

$$\bar{u}_i = 0 \quad \text{on } \partial R_v^g, \quad (3.3)$$

$$\ddot{u}_i = (\dot{u}'_i) + (\dot{u}_i)_{,j} c_j, \quad (3.4)$$

and τ_{ij} is given by eq. (3.1), $\rho > 0$ is the density, μ is the dynamic viscosity, b_i is the prescribed body force per unit volume R_v ; h_i is the prescribed tractions; and $i = 1, \dots, N_{sd}$, where N_{sd} is the space dimension.

4. Linearized equations of motion using Lagrangian description

A class of rate-type constitutive equations for inviscid materials has been introduced and discussed in [29,46], in which

$$\dot{\tau}_{ij} = C_{ijkl} \dot{u}_{(k,l)} + S_{ijkl} \dot{u}_{[k,l]}, \quad (4.1)$$

and where

$$C_{ijkl} = C_{ijkl} + C_{ijkl}^*. \quad (4.2)$$

Its corresponding linearized form is given as:

$$\Delta \tau_{ij} = \bar{C}_{ijkl} \Delta u_{(k,l)} + S_{ijkl} \Delta u_{[k,l]}, \quad (4.3)$$

where $\Delta u_{(k,l)}$ and $\Delta u_{[k,l]}$ are the symmetric and skew-

symmetric parts of the incremental displacement respectively.

Following techniques in [26,27,48] for deriving linearized variational equations, and employing eq. (4.3) the linearized version of eq. (3.2) is given as follows:

$$\begin{aligned} \int_{R_y} \rho \Delta \ddot{u}_i \bar{u}_i dR_y + \int_{R_y} \bar{u}_{i,j} D_{ijkl} \Delta u_{k,l} dR_y = \\ = \text{external force,} \\ - \text{internal force,} \\ - \text{inertial force,} \end{aligned} \quad (4.4)$$

where

$$D_{ijkl} = C_{ijkl} + \tau_{jl} \delta_{ik}, \quad (4.5)$$

$$\text{external force} = \int_{R_y} b_i \bar{u}_i dR_y + \int_{\partial R_y^h} h_i \bar{u}_i dA_y, \quad (4.6)$$

$$\text{internal force} = \int_{R_y} \tau_{ij} \bar{u}_{(i,j)} dR_y, \quad (4.7)$$

$$\text{inertial force} = \int_{R_y} \rho \ddot{u}_i \bar{u}_i dR_y. \quad (4.8)$$

The linearized constitutive equation (4.3) is used to derive the tangential initial stress matrix, but an improved representation can be used to integrate the constitutive equation (see e.g. [29],[32]).

Selective/reduced integration procedures [29,46] are employed to compute $\Delta u_{(k,l)}$ and $\Delta u_{[k,l]}$. The incremental constitutive equation (4.3) is a linearized expression. In a numerical formulation, the stresses will be calculated at all Gauss quadrature points which correspond to a set of material particles. The first term on the right-hand side of the equations represents the material response due to deformations, whereas the second term accounts for rotational effects, which are uniquely specified by ‘objectivity’. However, standard time-discretization procedures [42–44,48] when applied to this incremental constitutive equation typically only achieve objectivity in the limit of vanishingly small time steps, which is not economical in practical computing. If one employs the polar decomposition to separate deformation and rotation effects, much larger time steps can be accommodated. However, it is too time consuming to extract the eigenvalues and eigenvectors at each Gauss point.

Hallquist [17] has shown, in the context of two-dimensional analysis, how to maintain objectivity for larger time steps. However, there are some awkward aspects of the implementation, and the generalization to three dimension is not apparent. Hughes [32] proposed

an improved algorithm for integrating rate constitutive equations in large-deformation analysis. The algorithm is fairly simply and was shown to be objective with respect to large rotation increments.

This technique [29,32] is employed to account for the rotational term (i.e., $S_{ijkl} \Delta u_{[k,l]}$) and the various standard numerical integration algorithms [2,45,48] to account for the material response part (i.e., $C_{ijkl} \Delta u_{(k,l)}$).

5. Matrix equations of the mixed Lagrangian–Eulerian variational equations

The standard Galerkin/finite element discretization procedures are applied to eqs. (3.2), (3.4), (4.4) and (4.5). Let N_{umnp} be the number of discretized ‘nodal points’, N_{umel} is the total number of elements used, N_a is the ‘shape function’ associated with node a and N_{en} is the number of nodes per element. The matrix counterparts of the combined variation equations ((3.2), (3.4), (4.4) and (4.5)) are summarized as follows:

$$\begin{aligned} \sum_{b=1}^{N_{en}} \sum_{e=1}^{N_{umel}} \bar{\mathbf{u}}^e [M_{ab} \Delta(\dot{\mathbf{u}}')_b^e + D \cdot N_{ab}^c \Delta \dot{\mathbf{u}}_b^e \\ + K_{ab}^d \Delta \mathbf{u}_b^e + K_{ab}^v \Delta \dot{\mathbf{u}}_b^e] = \\ = \mathbf{F}^b + \mathbf{F}^s - \mathbf{F}^{\text{int}} - \mathbf{I} - \mathbf{N}^c, \end{aligned} \quad (5.1)$$

where

$$\mathbf{M} = [M_{ab}] = \int_{R_y^e} \rho N_a^t N_b dR_y, \quad (5.2)$$

$$\begin{aligned} D \cdot N^c &= [D \cdot N_{ab}^c] \\ &= \int_{R_y^e} (\mathbf{B}_a^t)^t [E^1(\mathbf{B}_b^t) + E^2(\mathbf{B}_b^t)] dR_y, \end{aligned} \quad (5.3)$$

$$\mathbf{K}^d = [K_{ab}^d] = \int_{R_y^e} \mathbf{B}_a^t \mathbf{D} \mathbf{B}_b dR_y, \quad (5.4)$$

$$\mathbf{K}^v = [K_{ab}^v] = \int_{R_y^e} \mathbf{B}_a^t \hat{\mathbf{D}} \mathbf{B}_b^t dR_y, \quad (5.5)$$

$$\mathbf{F}^b = [F_a^b] = \int_{R_y^e} N_a^t \mathbf{b} dR_y, \quad (5.6)$$

$$\mathbf{F}^s = [F_a^s] = \int_{\partial R_y^e} N_a^t \mathbf{h} dA_y, \quad (5.7)$$

$$\mathbf{F}^{\text{int}} = [F_a^{\text{int}}] = \int_{R_y^e} \mathbf{B}_a^t \boldsymbol{\tau} dR_y, \quad (5.8)$$

$$\mathbf{I} = [I_a] = \int_{R_y^e} \rho N_a^t (\dot{\mathbf{u}}') dR_y, \quad (5.9)$$

$$N^c = [N_a^c] = \int_{R_y} \rho N_a^i \boldsymbol{\eta} \, dR_y, \quad (5.10)$$

$$\mathbf{B}_a^1 = \begin{bmatrix} N_a & 0 & 0 \\ 0 & N_a & 0 \\ 0 & 0 & N_a \end{bmatrix} = \mathbf{N}_a, \quad (5.11)$$

$$\mathbf{E}^1 = \begin{bmatrix} \rho \dot{u}_{1,1} & \rho \dot{u}_{1,2} & \rho \dot{u}_{1,3} \\ \rho \dot{u}_{2,1} & \rho \dot{u}_{2,2} & \rho \dot{u}_{2,3} \\ \rho \dot{u}_{3,1} & \rho \dot{u}_{3,2} & \rho \dot{u}_{3,3} \end{bmatrix}, \quad (5.12)$$

$$\mathbf{B}_a^2 = \begin{bmatrix} N_{a,1} & 0 & 0 \\ N_{a,2} & 0 & 0 \\ N_{a,3} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & N_{a,1} & 0 \\ 0 & N_{a,2} & 0 \\ 0 & N_{a,3} & 0 \\ 0 & 0 & N_{a,1} \\ 0 & 0 & N_{a,2} \\ 0 & 0 & N_{a,3} \end{bmatrix}, \quad (5.13)$$

$$\mathbf{E}^2 = \rho \begin{bmatrix} c_1 & c_2 & c_3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & c_1 & c_2 & c_3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & c_1 & c_2 & c_3 \end{bmatrix}, \quad (5.14)$$

$$\mathbf{B}_a = \begin{bmatrix} \mathbf{B}_a^\gamma \\ \mathbf{B}_a^\theta \end{bmatrix}, \quad (5.15)$$

$$\mathbf{D} = \begin{bmatrix} \mathbf{C} & \mathbf{0}_{63} \\ \mathbf{0}_{36} & \mathbf{0}_{33} \end{bmatrix} + \mathbf{T}, \quad (5.16)$$

$$\hat{\mathbf{D}} = \begin{bmatrix} \Delta t \rho \beta + 2\mu & 0 & \Delta t \rho \beta & \Delta t \rho \beta & 0 & 0 \\ 0 & \mu & 0 & 0 & 0 & 0 \\ \Delta t \rho \beta & 0 & \Delta t \rho \beta + 2\mu & \Delta t \rho \beta & 0 & 0 \\ \Delta t \rho \beta & 0 & \Delta t \rho \beta & \Delta t \rho \beta + 2\mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{bmatrix}, \quad (5.17)$$

$$\mathbf{B}_a^\gamma = \begin{bmatrix} N_{a,1} & 0 & 0 \\ N_{a,2} & N_{a,1} & 0 \\ 0 & N_{a,2} & 0 \\ 0 & 0 & N_{a,2} \\ 0 & N_{a,3} & N_{a,2} \\ N_{a,3} & 0 & N_{a,1} \end{bmatrix}, \quad (5.18)$$

$$\mathbf{B}_a^\theta = \begin{bmatrix} N_{a,2} & -N_{a,1} & 0 \\ 0 & N_{a,3} & -N_{a,2} \\ -N_{a,3} & 0 & N_{a,1} \end{bmatrix}, \quad (5.19)$$

$$\boldsymbol{\eta} = \begin{bmatrix} c_1 \dot{u}_{1,1} + c_2 \dot{u}_{1,2} + c_3 \dot{u}_{1,3} \\ c_1 \dot{u}_{2,1} + c_2 \dot{u}_{2,2} + c_3 \dot{u}_{2,3} \\ c_1 \dot{u}_{3,1} + c_2 \dot{u}_{3,2} + c_3 \dot{u}_{3,3} \end{bmatrix}, \quad (5.20)$$

where $\mathbf{0}_{mn}$ denotes the $m \times n$ zero matrix, $\mathbf{C}$ is a 6×6 matrix of material tangent moduli whose components

Table 1
Initial-stress matrix

$$\mathbf{T} = \begin{bmatrix} \tau_1 & \frac{\tau_2}{2} & 0 & 0 & 0 & \frac{\tau_6}{2} & \frac{\tau_2}{2} & 0 & -\frac{\tau_6}{2} \\ & \frac{\tau_1 + \tau_3}{4} & \frac{\tau_2}{2} & 0 & \frac{\tau_6}{4} & \frac{\tau_5}{4} & \frac{\tau_3 - \tau_1}{4} & \frac{\tau_6}{4} & -\frac{\tau_5}{4} \\ & & \tau_3 & 0 & \frac{\tau_5}{4} & 0 & -\frac{\tau_2}{2} & \frac{\tau_5}{2} & 0 \\ \hline & & & \tau_4 & \frac{\tau_5}{2} & \frac{\tau_6}{2} & 0 & -\frac{\tau_5}{2} & \frac{\tau_6}{2} \\ & & & & \frac{\tau_3 + \tau_4}{4} & \frac{\tau_2}{4} & -\frac{\tau_6}{4} & \frac{\tau_4 - \tau_3}{4} & \frac{\tau_2}{4} \\ & & & & & \frac{\tau_4 + \tau_1}{4} & \frac{\tau_5}{4} & -\frac{\tau_2}{4} & \frac{\tau_1 - \tau_4}{4} \\ \hline \text{Symmetric} & & & & & & \frac{\tau_1 + \tau_3}{4} & -\frac{\tau_6}{4} & -\frac{\tau_5}{4} \\ & & & & & & & \frac{\tau_3 + \tau_4}{4} & -\frac{\tau_2}{4} \\ & & & & & & & & \frac{\tau_4 + \tau_1}{4} \end{bmatrix} \quad (5.22)$$

are given by $C_{IJ} = C_{ijkl}$ in which the relationships between the indices are given by:

$I \backslash J$	$i \backslash k$	$J \backslash l$
1	1	1
2	1	2
3	2	2
4	3	3
5	2	3
6	3	1

(5.21)

The initial-stress matrix, T , is defined in table 1, where

$$\tau_I = \tau_{ij} \quad (5.23)$$

The D matrix is arranged to be compatible with the following ordering of strain and rotation components:

$$\gamma_{11}, 2\gamma_{12}, \gamma_{22}, \gamma_{33}, 2\gamma_{23}, 2\gamma_{31}, 2\omega_{12}, 2\omega_{23}, 2\omega_{31}, \quad (5.24)$$

in which

$$\gamma_{ij} = (\Delta u_{i,j} + \Delta u_{j,i})/2$$

and

$$\omega_{ij} = (\Delta u_{i,j} - \Delta u_{j,i})/2. \quad (5.25)$$

6. 'Upwind' finite elements by modifying the standard Gauss-Legendre rules

As can be seen from eq. (5.1), the second term constitutes a nonsymmetric operator. In structural and solid mechanics, the operator is a symmetric one. Until recently, nonself-adjoint cases (nonsymmetric operators) have not been well understood. If Gauss quadrature is used on the convective term appearing in the above-mentioned equation, spurious spatial oscillations occur for some high Reynolds number flows. These noisy results can only be corrected by mesh refinement. However, in most problems of engineering interest, convection is dominant and the geometry is so complex that severe mesh refinement is not economically feasible. To circumvent these problems, 'upwind' finite elements [8,25,30,31,33] or Petrov-Galerkin methods [9,16,20,33] are used as alternatives to generate stable finite element approximations to highly convective flows.

The method proposed in [30,31] is based upon numerical integration rules which are simple, efficient, very easy to implement, and yet effective. Instead of using the standard Gauss-Legendre rules, a modified

'upwind' integration rule was employed. The positions of the integration points are determined from the 'element Reynolds number' (see e.g., [30,31]). The performance of the 'upwind' scheme used in [30,31] was found to be satisfactory for a wide class of flow problems. However, for some time-dependent convection-dominated flow problems it was found that the 'spurious crosswind diffusion' produced by the upwind scheme degrades the accuracy of the solution (see e.g., [8,25]). A multidimensional upwind scheme with no 'crosswind diffusion' was designed and shown to be superior to the quadrature scheme [8,25].

It is still too early to evaluate the relative merits of the various schemes. Although these schemes differ considerably in concept and implementation, they are 'designed' to achieve similar ends.

The upwind scheme with no 'crosswind diffusion' proposed in [8,25] will be used. However, the scheme [8,25], which is restricted to the Eulerian case, has to be modified for mixed Eulerian-Lagrangian formulation. Following [8,25], the equation of motion in the mixed Lagrangian-Eulerian description is modified to:

$$\rho(\dot{u}'_i) + \rho c_j(\dot{u}_i)_{,j} = (\tau_{ij} + D_{ij})_{,j}, \quad (6.1)$$

where D_{ij} is an artificial viscous force and is determined by:

$$D_{ij} = \dot{u}_{i,k} \tilde{\mu}_{jk}, \quad (6.2)$$

in which

$$\tilde{\mu}_{jk} = \tilde{\mu} \hat{c}_k \hat{c}_j, \quad (6.3)$$

$$\hat{c}_k = c_k / \|c\|, \quad (6.4)$$

and $\tilde{\mu}$ is defined as follows:

$$\tilde{\mu} = \rho \left(\sum_{i=1}^{N_{sd}} \tilde{\xi}_i c_{\xi_i} h_{\xi_i} \right) / N_{sd}, \quad (6.5)$$

where

$$\xi_i = \coth \alpha_{\xi_i} - 1/\alpha_{\xi_i} \quad (6.6)$$

or approximately

$$\xi_i = \begin{cases} \alpha_{\xi_i}/3 & -3 \leq \alpha_{\xi_i} \leq 3, \\ \text{sgn } \alpha_{\xi_i} & |\alpha_{\xi_i}| > 3, \end{cases} \quad (6.7)$$

in which

$$\alpha_{\xi_i} = \rho c_{\xi_i} h_{\xi_i} / 2\mu \quad (6.8)$$

and

$$c_{\xi_i} = e_{\xi_i} \cdot c. \quad (6.9)$$

The meaning of $h_{\xi i}$ and $e_{\xi i}$ is illustrated in fig. 1 for the four-node bilinear quadrilateral. The corresponding variational equation for (3.10) can be written as follows:

$$\begin{aligned} & \int_{R_y} [\rho \bar{u}_i (\dot{u}_i') + \tau_{ij} \bar{u}_{i,j}] dR_y \\ & + \int_{R_y} \rho (\bar{u}_i + \tilde{u}_i) c_{ij} u_{i,j} dR_y = \\ & = \int_{R_y} b_i \bar{u}_i dR_y + \int_{\partial R_y^h} h_i \bar{u}_i dA_y, \end{aligned} \quad (6.10)$$

in which

$$\bar{u}_i = 0 \quad \text{on } \partial R_y^g \quad (6.11)$$

and

$$\tilde{u}_i = \frac{\tilde{\mu}}{\rho \|c\|} \bar{u}_{i,k} \hat{c}_k. \quad (6.12)$$

In [8], a 'consistent weighting' is suggested for the intended applications. It was thought to be sufficient at this stage of development to neglect consistent weighting. Nevertheless, it is anticipated that a consistent weighted formulation (Petrov-Galerkin scheme) would lead to improved results in transient analysis.

7. Shell element and contact/sliding element

7.1. Shell element

The equations of motion derived in previous sections can be used directly in shell analysis. The Lagrangian formulation will be adopted. The shell element used in the one which is based upon Mindlin theory [49] in which it can accommodate finite strains and finite rotations, and permits use of arbitrary, three-dimensional nonlinear constitutive equations (see e.g. Hughes and Liu [29], Liu [46]).

7.2. Contact/sliding element

Contact elements may be used to impose inequality constraints between nodes. Either perfect friction (i.e., 'stick') or frictionless (i.e., 'slide') conditions may be achieved.

A contact element is defined by three nodes and a spring constant, or 'penalty parameter', k , for our purposes. The connection from node A to node B defines the 'slide line' direction; whereas, node C is the contact node (see fig. 2). The normalized projected distance of node C to node A is denoted by $\alpha \in [0,1]$ and is given

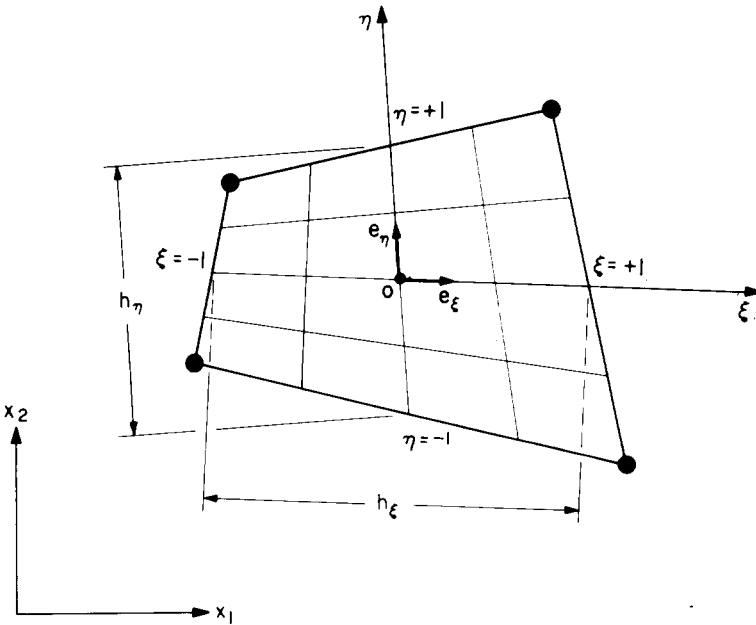


Fig. 1. Typical four-node quadrilateral finite element geometry.

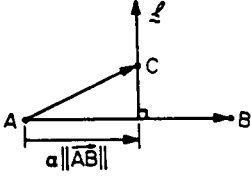


Fig. 2. A contact element is defined by three nodes.

by:

$$\alpha = \frac{\vec{AC} \cdot \vec{AB}}{\|\vec{AB}\|^2}. \quad (7.1)$$

Here $\vec{XY}$ denotes a vector connecting point X to point Y and ' $\cdot$ ' is the dot product. The normal vector l from node C to $\vec{AB}$ is defined by:

$$l = \left(\vec{AB} \times \vec{AC} \right) \times \vec{AB}, \quad (7.2)$$

$$l \leftarrow l / \|l\|, \quad (7.3)$$

where ' $\times$ ' is the cross product and ' $\leftarrow$ ' means 'is replaced by'. The local contact stiffness matrix K^l is:

$$K^l = k \begin{bmatrix} (1-\alpha)^2 & \alpha(1-\alpha) & -(1-\alpha) \\ \text{sym} & \alpha^2 & -\alpha \\ & & 1 \end{bmatrix}. \quad (7.4)$$

The rows and columns of the contact stiffness matrix are arranged in such a way that the first row or column corresponds to node A, while the 2nd row (or column) and 3rd row (or column) correspond to nodes B and C, respectively. Before the global assembly procedure, the local contact stiffness matrix is rotated to the global

form K^G (for the sliding case) via:

$$K^G = R^l K^l R, \quad (7.5)$$

$$R = \begin{bmatrix} I^t & \mathbf{0}_{13} & \mathbf{0}_{13} \\ \mathbf{0}_{13} & I^t & \mathbf{0}_{13} \\ \mathbf{0}_{13} & \mathbf{0}_{13} & I^t \end{bmatrix}, \quad (7.6)$$

$$\mathbf{0}_{13} = [0 \quad 0 \quad 0] \quad (7.7)$$

For the sticking case K^G is to find in table 2.

8. Implicit-explicit temporal integration of the variational equations

Background information on implicit-explicit temporal integration may be found in [4-6,34-37,39]. A general partitioned transient analysis procedure is proposed by Park [52], which is amenable to a unified stability analysis. Various other improvements in implicit-explicit transient algorithms have also been recently achieved [38,46,51,53,54,58,63].

The new family of implicit-explicit algorithms developed by Hughes and Liu [34,35] is employed for the temporal integration of the equations. Although there are many difference formulas that can be used in time discretization, based on previous studies, the Newmark family of methods [50] appears quite suitable.

Let $\hat{d}_n$, $\hat{v}_n$, and $\hat{a}_n$ be the nodal vectors of mesh displacements, mesh velocities, and mesh accelerations (i.e., $\hat{u}_n$, $\hat{u}'_n$, $\hat{u}''_n$) respectively; likewise, d_n , v_n , and a_n are the material displacements, material velocities and material referential accelerations (i.e., u_n , u'_n , u''_n), respectively. The solution at time step $n+1$ is determined by the following equations (in order):

$$i = 0 \quad (i \text{ is the iteration counter}). \quad (8.1)$$

Table 2

$$K^G = \begin{bmatrix} (1-\alpha)^2 & 0 & 0 & \alpha(1-\alpha) & 0 & 0 & -(1-\alpha) & 0 & 0 \\ & (1-\alpha)^2 & 0 & 0 & \alpha(1-\alpha) & 0 & 0 & -(1-\alpha) & 0 \\ & & (1-\alpha)^2 & 0 & 0 & \alpha(1-\alpha) & 0 & 0 & (1-\alpha) \\ & & & \alpha^2 & 0 & 0 & -\alpha & 0 & 0 \\ \text{Symmetric} & & & & \alpha^2 & 0 & 0 & -\alpha & 0 \\ & & & & & \alpha^2 & 0 & 0 & -\alpha \\ & & & & & & 1 & 0 & 0 \\ & & & & & & & 1 & 0 \\ & & & & & & & & 1 \end{bmatrix} \quad (7.8)$$

Predictor phase for material displacements, velocities, and accelerations:

$$\mathbf{d}_{n+1}^{(i)} = \tilde{\mathbf{d}}_{n+1}, \quad (8.2)$$

$$\mathbf{v}_{n+1}^{(i)} = \tilde{\mathbf{v}}_{n+1}, \quad (8.3)$$

$$\mathbf{a}_{n+1}^{(i)} = \mathbf{0}, \quad (8.4)$$

$$\tilde{\mathbf{d}}_{N+1} = \mathbf{d}_n + \Delta t \mathbf{v}_n + \left(\frac{1}{2} - \beta\right) \Delta t^2 \mathbf{a}_n, \quad (8.5)$$

$$\tilde{\mathbf{v}}_{n+1} = \mathbf{v}_n + (1 - \gamma) \Delta t \mathbf{a}_n. \quad (8.6)$$

Predictor phase for mesh displacements, and velocities:

$$\tilde{\mathbf{d}}_{n+1}^{(i)} = \tilde{\mathbf{d}}_{n+1}, \quad (8.7)$$

$$\tilde{\mathbf{v}}_{n+1}^{(i)} = \tilde{\mathbf{v}}_{n+1}, \quad (8.8)$$

$$\tilde{\mathbf{d}}_{n+1} = \hat{\mathbf{d}}_n + \Delta t \hat{\mathbf{v}}_n + \left(\frac{1}{2} - \beta\right) \Delta t^2 \hat{\mathbf{a}}_n, \quad (8.9)$$

$$\tilde{\mathbf{v}}_{n+1} = \hat{\mathbf{v}}_n + (1 - \gamma) \Delta t \hat{\mathbf{a}}_n. \quad (8.10)$$

Momentum equations:

$$\mathbf{M}^* \Delta \mathbf{a} = \Delta \mathbf{F}^*, \quad (8.11)$$

$$\mathbf{M}^* = \mathbf{M} + \gamma \Delta t \mathbf{D} \cdot \mathbf{N}^c + \gamma \Delta t \mathbf{K}^v + \beta \Delta t^2 \mathbf{K}^d, \quad (8.12)$$

$$\Delta \mathbf{F}^* = \mathbf{F}^{\text{ext}} + \mathbf{F}^b + \mathbf{F}^s - \mathbf{I} - \mathbf{N}^c - \mathbf{F}^{\text{int}}, \quad (8.13)$$

$$\mathbf{F}^{\text{ext}} = \text{discrete nodal applied forces.} \quad (8.14)$$

Corrector phase for material displacements, velocities, and accelerations:

$$\mathbf{a}_{n+1}^{(i+1)} = \mathbf{a}_{n+1}^{(i)} + \Delta \mathbf{a}, \quad (8.15)$$

$$\mathbf{v}_{n+1}^{(i+1)} = \tilde{\mathbf{v}}_{n+1} + \gamma \Delta t \mathbf{a}_{n+1}^{(i+1)}, \quad (8.16)$$

$$\mathbf{d}_{n+1}^{(i+1)} = \tilde{\mathbf{d}}_{n+1} + \beta \Delta t^2 \mathbf{a}_{n+1}^{(i+1)}. \quad (8.17)$$

Mesh velocity representation:

$$\hat{\mathbf{v}}_{n+1}^{(i+1)} = \mathbf{A}(\mathbf{v}_{n+1}^{(i+1)} - \mathbf{L}^{-1} \mathbf{S}). \quad (8.18)$$

Corrector phase for mesh accelerations and displacements:

$$\hat{\mathbf{a}}_{n+1}^{(i+1)} = (\hat{\mathbf{v}}_{n+1}^{(i+1)} - \tilde{\mathbf{v}}_{n+1}) / \gamma \Delta t, \quad (8.19)$$

$$\hat{\mathbf{d}}_{n+1}^{(i+1)} = \tilde{\mathbf{d}}_{n+1} + \beta \Delta t^2 \hat{\mathbf{a}}_{n+1}^{(i+1)}. \quad (8.20)$$

In eq. (8.18), we recall from [31] that $\mathbf{A}$ is the Lagrange-Euler parameter (a diagonal matrix). $\mathbf{L}$ and $\mathbf{S}$ are defined as follows (see [31] for details):

$$\mathbf{L} = \sum_{e=1}^{\text{Numel}} \int_{R_x^e} \mathbf{N}^t \mathbf{N} R_x, \quad (8.21)$$

$$\mathbf{S} = \sum_{e=1}^{\text{Numel}} \int_{R_x^e} \sum_{a=1}^{N_{en}} N_a \mathbf{c}_k^e a \left(\sum_{b=1}^{N_{en}} N_{b,k} \times (\hat{\mathbf{u}}_b^e + \hat{\mathbf{x}}_b^e) \right) dR_x. \quad (8.22)$$

Let I be the total number of iterations. If $I = 0$, the algorithm is a predictor-corrector algorithm. If more iterations are required (i.e., $I > 0$), it is a predictor-multicorrector algorithm. For static analysis, the algorithm may be reduced to a standard Newton-Raphson iterative procedure by setting $\gamma = 0, \beta \Delta t^2 = 1, \tilde{\mathbf{d}}_{n+1} = \mathbf{d}_n$, and ignoring all equations in which the velocity appears.

The generalizations of eqs. (8.1)–(8.20) to implicit, explicit, implicit-explicit mesh partitions, implicit-explicit operator splitting, dynamic relaxation, and quasi-Newton updates for symmetric and non-symmetric equation systems may be done by modifying $\mathbf{M}^*$ and $\Delta \mathbf{F}^*$ accordingly (see [35,36,39] for details).

9. Application to fluid-structure interaction

As a practical application, the finite element procedures described here are employed for the dynamic, three-dimensional, nonlinear, inelastic response of ground-supported, cylindrical liquid-filled tanks. Both fluid-structure interaction and free-surface sloshing effects are treated.

Early studies relevant to this topic were performed by Westergaard [62], Hoskins and Jacobsen [21], Jacobsen [41], Werner and Sundquist [61], and Jacobsen and Ayre [40]. They assumed the tank to be rigid and then determined the hydrodynamic wall pressures. Housner [22] developed an analytical method for determining hydrodynamic wall pressure under the same assumption that the tank is rigid. A mechanical equivalent model is developed and is widely used in current earthquake engineering practice [23].

The subject appears to have been laid to rest until the 1964 Alaska earthquake, which caused large-scale damage to tanks of modern design [18,55] for the first time and greatly influenced the research into their dynamic characteristics.

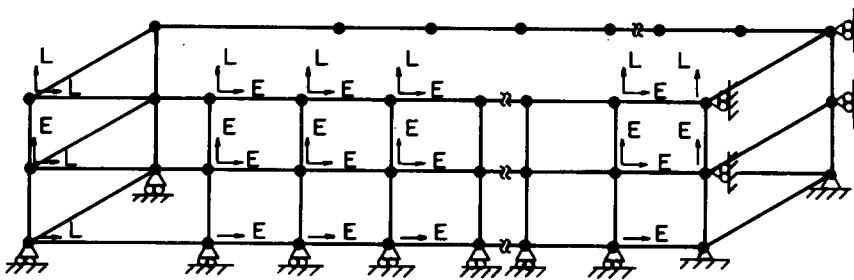
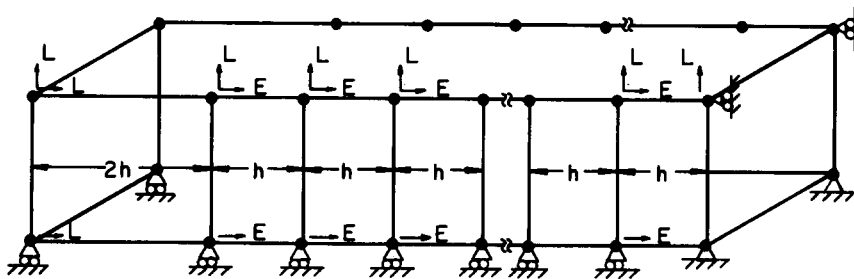
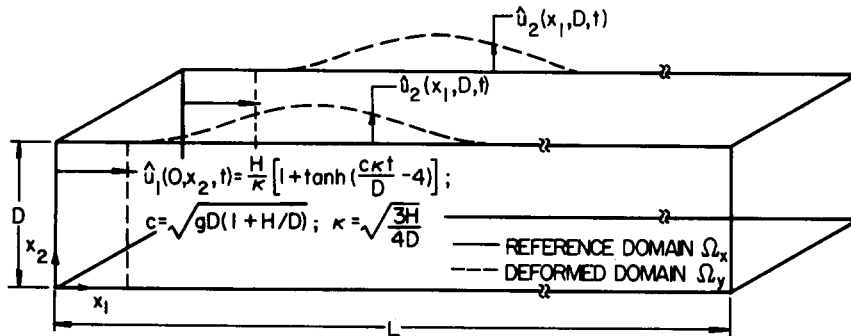
Veletsos [59], Yang [65], and Veletsos and Yang [60] have shown, however, that the seismic effects in flexible tanks may be substantially greater than those induced in similarly excited rigid tanks. The first use of a digital computer in analyzing the problem with the aid of finite element method was completed in 1969 by Edwards [13]. This investigation treated the coupled interaction

between the elastic wall of the tank and the contained liquid. Recent developed along the same lines were performed by Wu et al. [64] and Shaaban and Nash [56].

Haroun [19] recently conducted a theoretical and experimental investigation of the dynamic behavior of cylindrical liquid storage tanks. His goal was the improvement in the design of these tanks to resist earthquakes. The natural frequencies of vibration were

found through the use of a scheme in which the elastic shell is modelled by finite elements and the fluid region is treated as a continuum by a boundary integral techniques. Based upon these studies, Housner and Haroun [24] proposed an improved design-procedure for liquid storage tanks.

However, the method cited are based on small motion linear response, and do not take satisfactory account



Key { L - Lagrangian degree-of-freedom
E - Eulerian degree-of-freedom

Fig. 3. Problem statement and finite-element meshes for free-surface flow.

of nonlinear effects observed in the field under the action of strong earthquakes, such as lift-off of tanks from foundations, finite-amplitude sloshing, and nonlinear and inelastic tank response. A more advanced approach which includes an analytical treatment of these nonlinear effects may provide a better understanding of the structural response and its relationship to design procedures.

9.1. Numerical examples

9.1.1. Free-surface wave-propagation problem

Since the capability of the method to propagate a wave is relevant to its applicability to a sloshing problem, we first consider a three-dimensional problem in which a surface wave moves in one direction. The problem statement and finite element meshes are shown in fig. 3. A wave is generated numerically by prescribing a displacement-time history at the left-hand boundary of the domain. The prescribed function is designed in such a way that it is consistent with the experimental studies. Its function is to produce a solitary wave—one in which the nonlinear and dispersive effects are balanced so that the wave propagates without distortion.

Data used in the analysis are given as follows: $g = 1$, $L = 949.095$, $D = 10$, $H = 0.86$, $h = 5.895$, and $\Delta t = h/c = 1.7888$. Meshes 1 and 2 consist of 160 and 320, 8-node elements, respectively. The flow was assumed inviscid, $\rho = 1$ and $\Delta t \rho \beta = 2.7 \times 10^7$.

The finite element calculations employing a one-dimensional depth averaged theory [15] are compared with calculations for mesh 1, one element through the depth. The results are shown in fig. 4. The figure shows the results agree remarkably well, considering the two approaches are quite different. Notice that the rate of growth of the trough is a little greater with the present scheme than it is for the one-dimensional theory. Even though there is good agreement between the two theories, experimental results are not consistent with the previously obtained numerical results. The differences are: (1) the experimental waveform propagates somewhat faster and exhibits less amplitude decay; and (2) dispersion emanating from the back of the wave is not seen experimentally.

The reason for the discrepancy between the numerical and experimental results is that the variation of the velocity through the depth is not linear. Both the one-dimensional depth averaging theory and one element through the depth two-dimensional theory invoke this assumption. As can be seen in fig. 5, with two elements through the depth, the trough has been reduced and the relative wave height is essentially constant with propa-

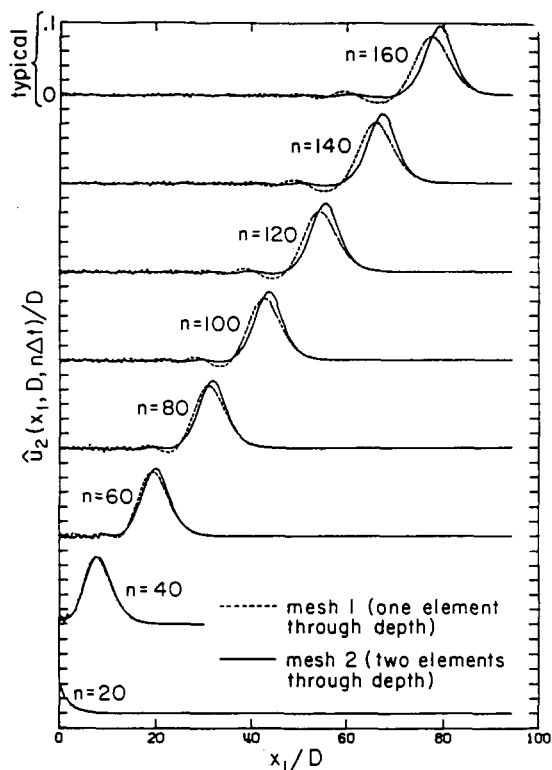


Fig. 4. Comparison of one-dimensional depth-averaged theory and two-dimensional Lagrangian-Eulerian formulation.

gation. The difference in having two instead of one element in depth is that the distribution of velocity is no longer constrained to be linear with depth.

It is also noted experimentally that 'noise' superposed upon a solitary wave tends to propagate at slower velocity than the main pulse. A numerical simulation of this phenomenon is shown in fig. 6. The noise is generated by setting a no-slip boundary condition along the left-hand edge of the domain, in contrast to previous calculations in which a slip condition was employed. Note how the solitary wave emerges, unaffected by the initial perturbations.

9.1.2. Cylindrical shell containing a fluid

A buckling analysis of a tilted cylindrical shell subjected to internal hydrostatic loading was performed. The problem statement, hoop stress, axial moment and axial stress results prior to buckling are shown in fig. 7. Data used in the analysis are given as follows: Young's modulus $E = 7.35 \times 10^5$; Poisson's ratio $\nu = 0.3$, $D =$

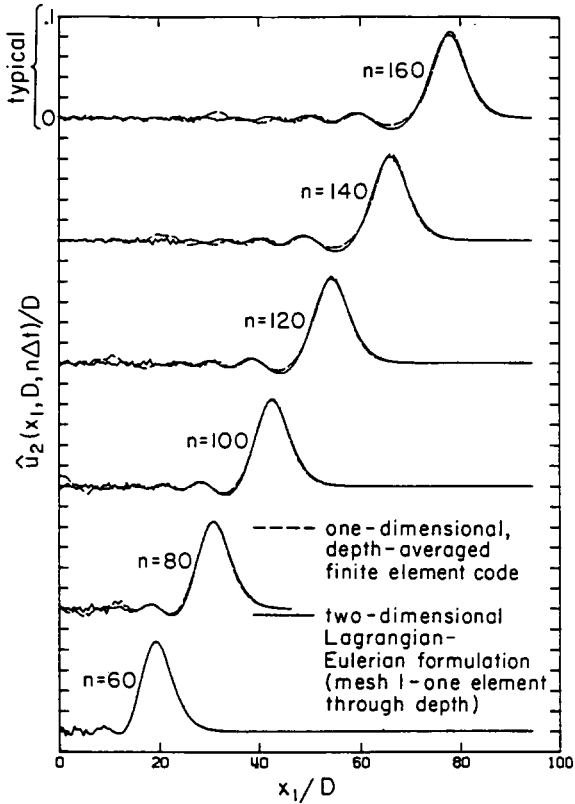


Fig. 5. Comparison of results for meshes 1 and 2.

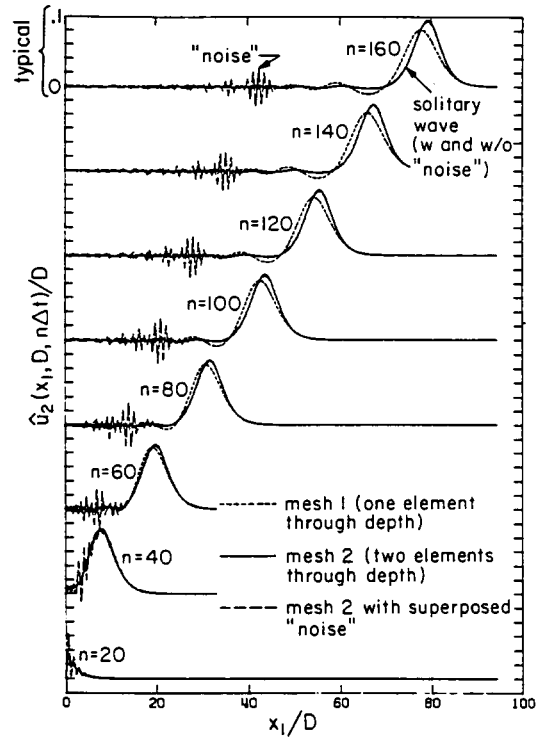


Fig. 6. Comparison of results with, and without, superposed 'noise'.

16.0, $L = 20.0$, $R = 4.0$, $h = 0.01$, $\alpha = 30.0$, and the density of the fluid $\gamma = 0.0361$.

In order to apply the hydrostatic load incrementally, the effective specific gravity, γ_{eff} , was defined by $\gamma_{\text{eff}} = \gamma_g$. The value of g used in obtaining the stresses shown in fig. 7 was 1.8. The correctness of the numerical results outside of the boundary layer at the fixed end is confirmed by comparison with a membrane theory solution [57]. Buckling calculations were performed by incrementing g . The deformed configurations and detail of lower region for the tilted cylindrical shell are shown in figs. 8 and 9, respectively. As may be seen, the initial bifurcation occurred in attempting to increment g beyond 1.8. This corresponds to a maximum axial membrane stress of 547.53 at $x = L$, $\phi = \pi$, which is approximately half the classical value. Although this result was not anticipated, it bears a great deal of similarity to results obtained by Argyris and Dunne [3] in their study of a compressed cylindrical panel. They too found an initial bifurcation of approximately half the classical

value. When they pursued the analysis into the post-buckled regime, a drop in load of only 3% was noted, at which time the load could be increased again to a value in the vicinity of classical buckling load. At this point a second, and more significant bifurcation ensued, and this was followed by several more bifurcations. An important point in considering the results of Argyris and Dunne was that a preliminary eigenvalue analysis, employing the same mesh used in the finite deformation case, yielded a buckling load approximately on the high side of the classical value. This lends credence to the analysis, but a full understanding of the computed initial bifurcation in this case, and the present case, does not yet seem to exist. The results shown were obtained using S1 elements (see e.g., [29]). Similar results were also obtained with heterosis elements (see e.g., [29]).

9.1.3. Dynamic analysis of a liquid filled cylindrical tank

A dynamic analysis was performed of a liquid filled

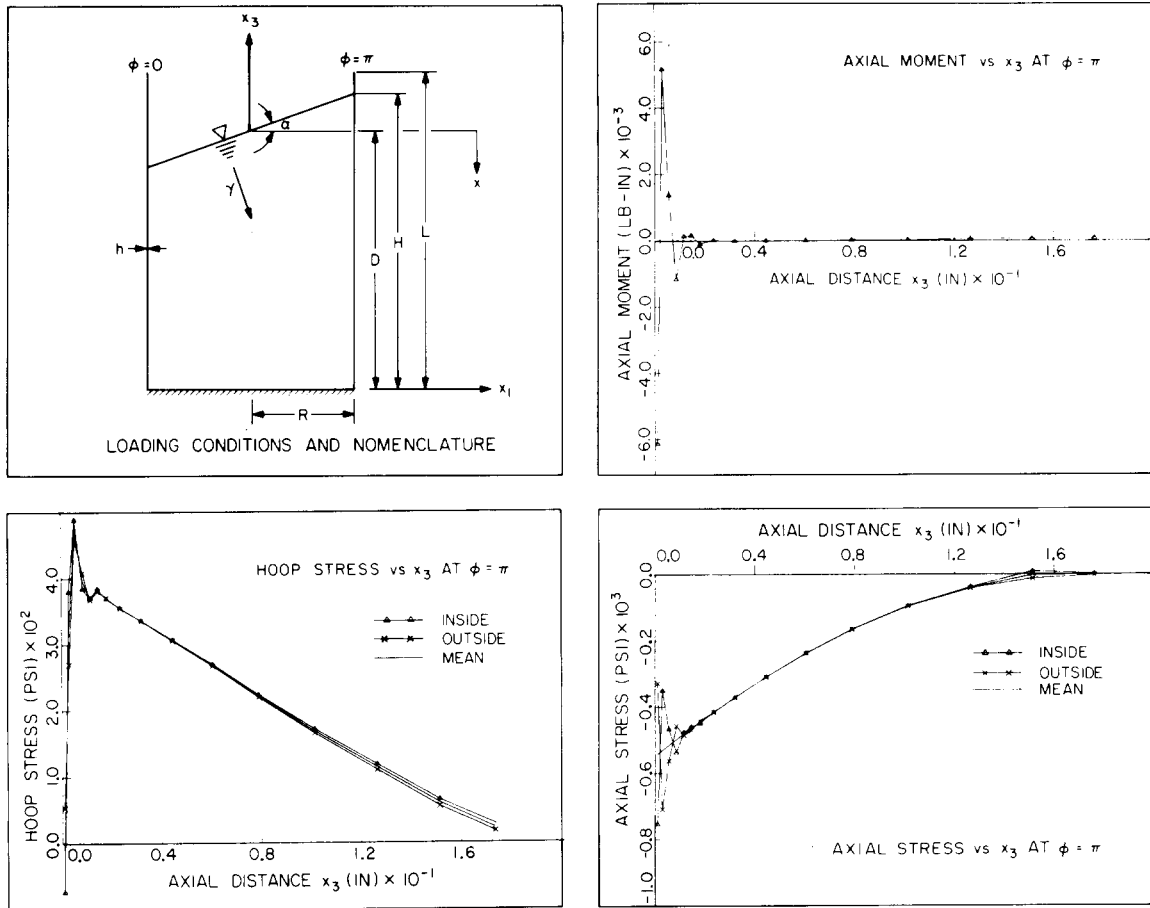


Fig. 7. Tilted cylindrical shell subjected to internal hydrostatic loading; stress results in the linear regime.

cylindrical tank subjected to a periodic saw-tooth ground acceleration excitation. The problem statement is shown in fig. 10, along with displacement, velocity, and acceleration time histories at three different positions (node A, B, and C). All the bottom nodes of the tank are subjected to the same excitation in the x -direction with the other two directions (y and z) fixed. A typical input ground excitation (acceleration, velocity, and displacement) is given for node A. Due to symmetry, only half the tank is modelled by 36 S1 elements, 80 uniform-reduced integration 8-node fluid elements, and 30 sliding elements. The height of the tank is 864 (i.e., distance between A and C). The mean diameter of the tank is 577 (i.e., distance between B and C). The tank thickness is 1. It is filled with liquid up to 720. The material properties of the tank and fluid are:

Tank

$E = 3.0 \times 10^7$, $\nu = 0.3$, $\rho = 7.324 \times 10^{-4}$, yield stress $\sigma_y = 3.0 \times 10^4$, and plastic modulus $B = 0.0$.

Fluid

Bulk modulus $\beta = 6.991 \times 10^{12}$, dynamic viscosity $\mu = 0.0$, $\rho = 9.349 \times 10^{-5}$, and gravity $g = 3.864 \times 10^2$.

The spring constant, k , of the contact elements is taken to be 4.44×10^{16} . The Newmark parameters are: $\beta = 0.5625$, $\gamma = 1.0$, and $\Delta t = 0.02$. A linear implicit-nonlinear explicit operator splitting algorithm with one interaction is employed. A small deformation analysis option is used. The maximum computed stress occurring in the tank is less than σ_y even though the peak ground excitation is 1.5g. The relative free surface motion as well as the relative tank displacement are found

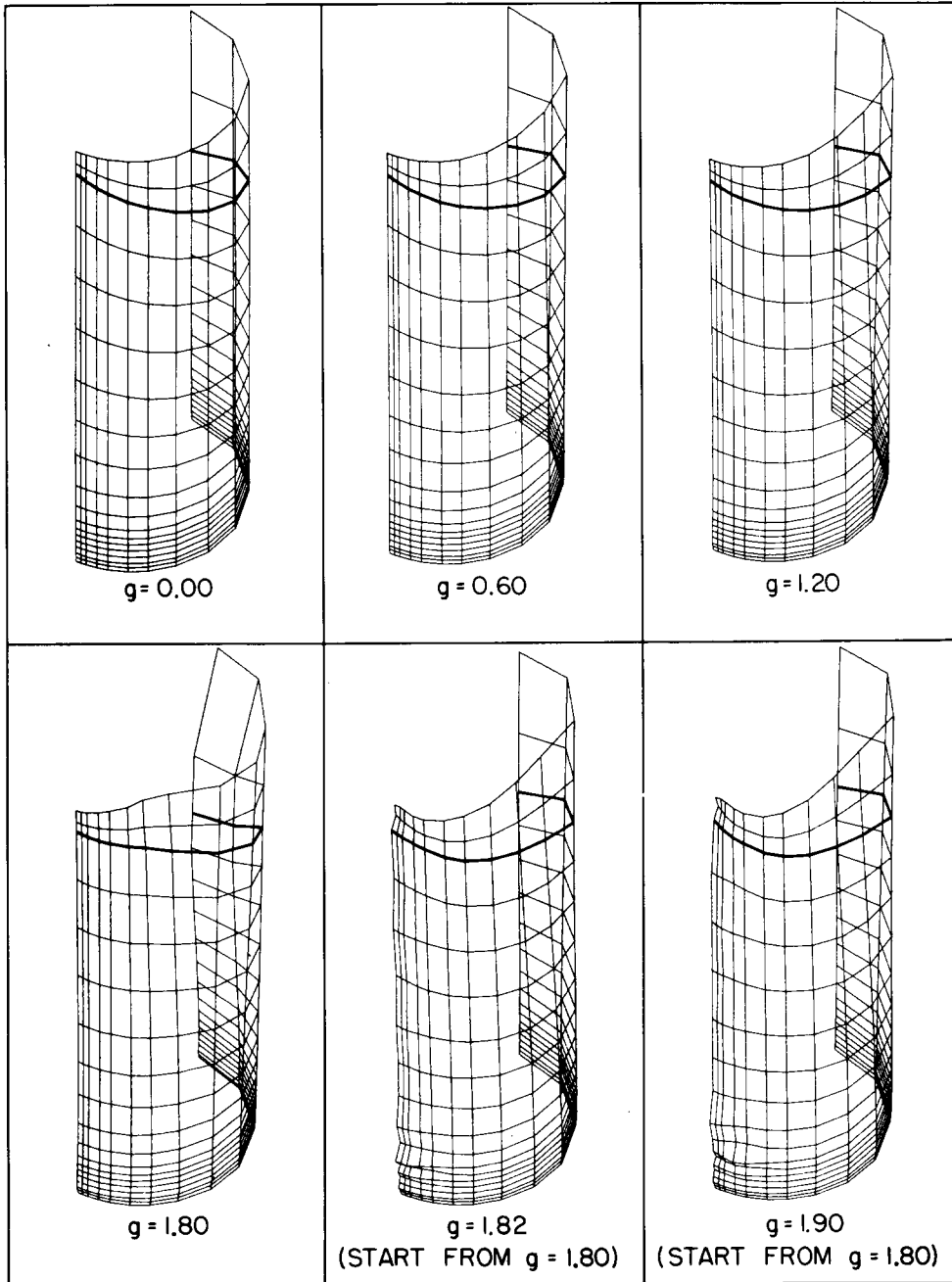


Fig. 8. Deformed configurations for the tilted cylindrical shell.

to be within the small motion region. Therefore, no further analysis of this problem was performed. The computed acceleration, velocity, and displacement time histories of nodes B and C as well as the relative

displacement between nodes B and A and between nodes C and A are shown in fig. 10. It is expected that the fundamental mode is dominant for this kind of excitation; also nodes B and C are expected to respond

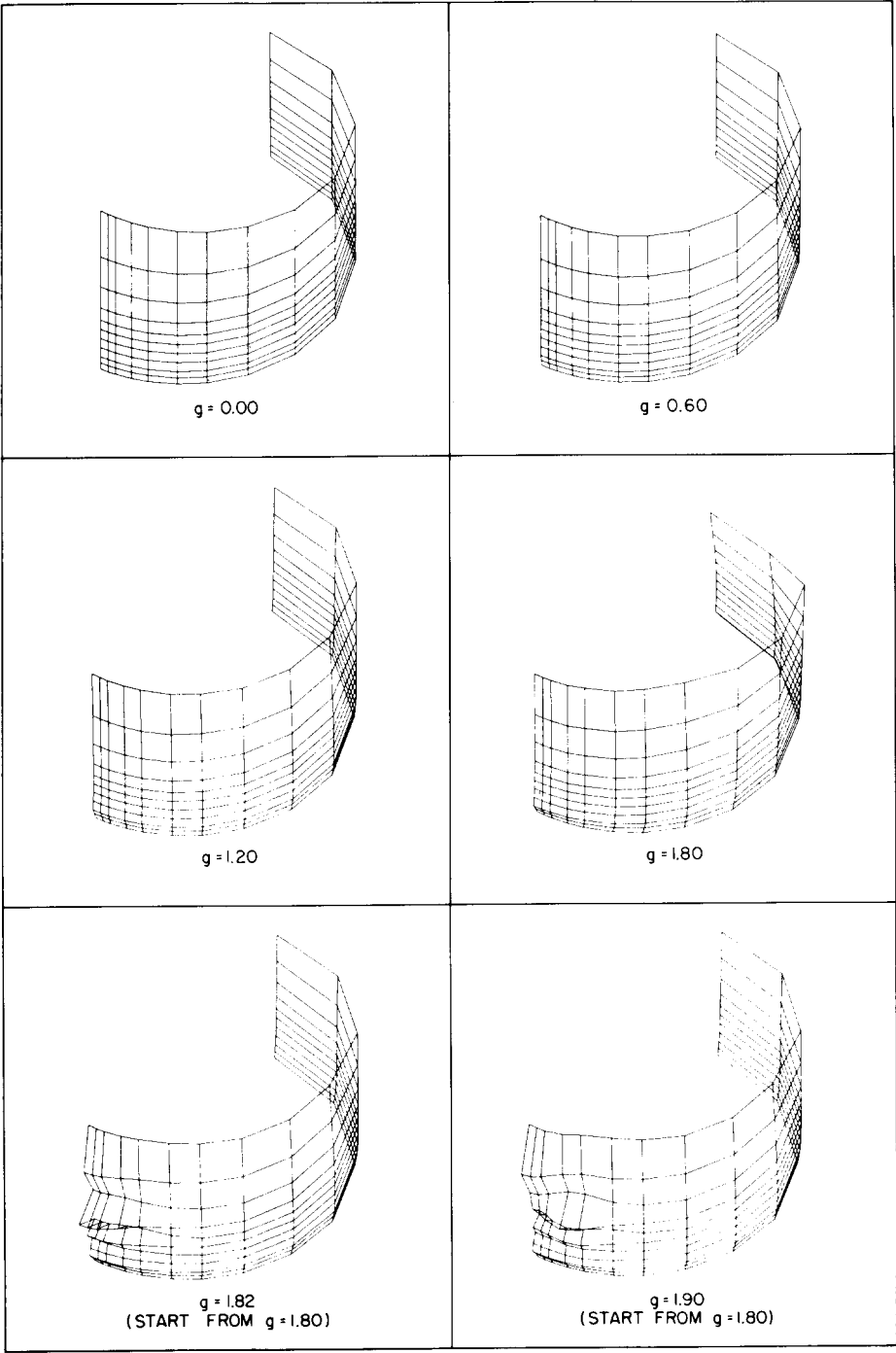


Fig. 9. Detail of lower regions for the tilted cylindrical shell.

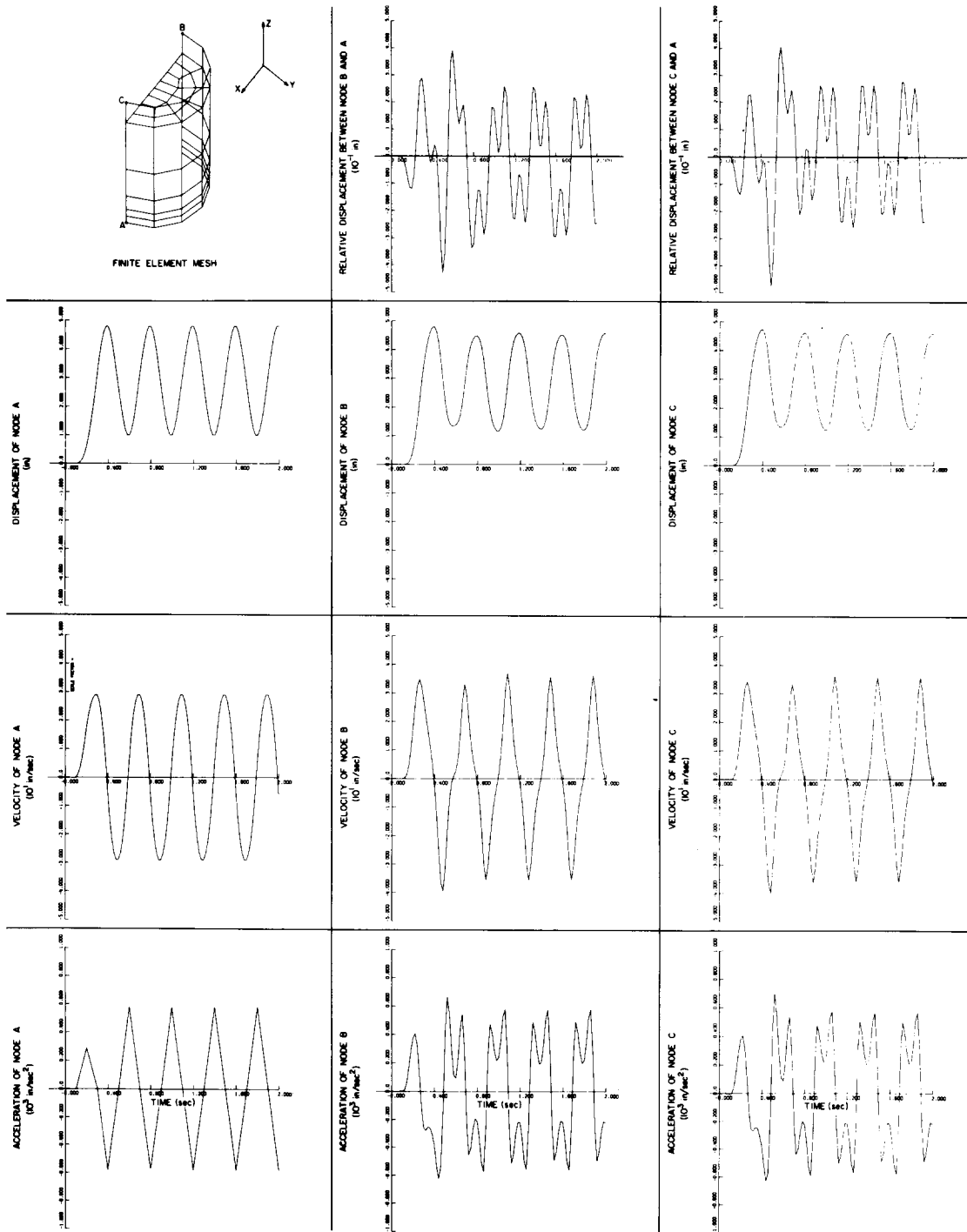


Fig. 10. Problem statement and displacement, velocity, and acceleration time histories at three different positions (node A, B, and C).

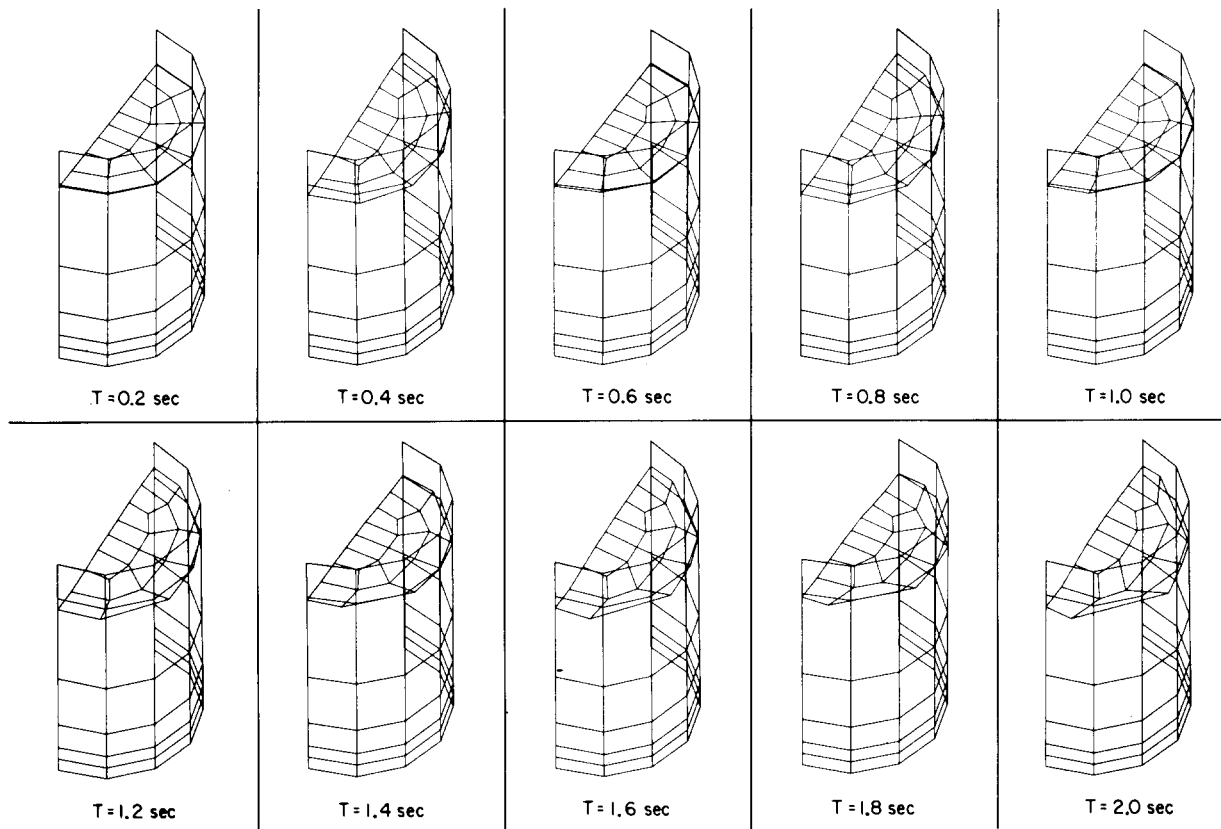


Fig. 11. Deformed shape of the tank and the free surface motion (with the input translational motion subtracted), magnified by a factor of 7.81.

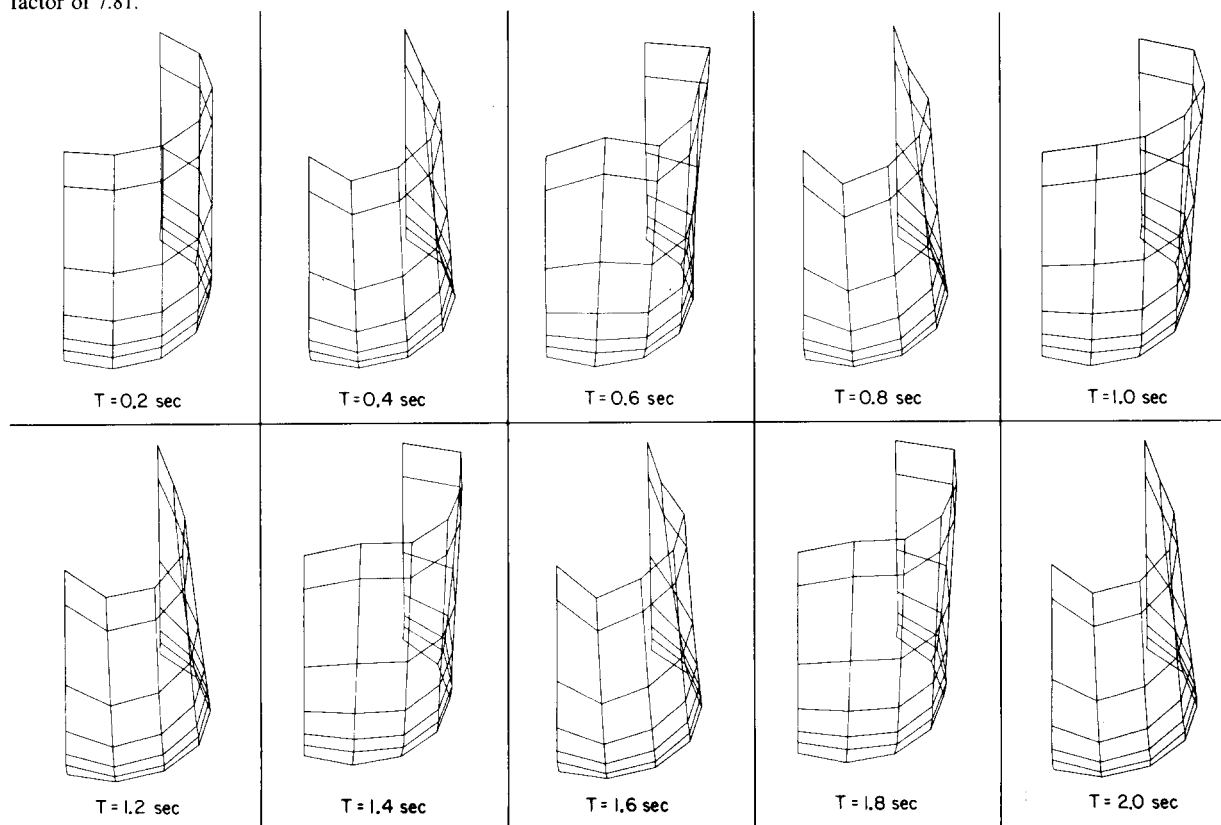


Fig. 12. Tank response (with the input translational motion subtracted), magnified by a factor of 76.31.

in 'phase'. This phenomenon can be seen from the absolute displacement-time histories of nodes B and C in the figure.

The deformed shape of the tank and the free surface motion (with the input translational motion subtracted) are shown in fig. 11 at a time interval of 0.2 seconds. The magnification factor of these plots is 7.81. Frequency responses higher than the fundamental sloshing frequency were observed. The sloshing period is approximately 6 seconds. Since the free surface motion is much larger than tank deformation, plots of the tank response (with the input translational motion subtracted) are shown in fig. 12, magnified by a factor of 76.31.

10. Conclusions

In this paper, finite element procedures for fluid-structure interaction problems with particular reference to the dynamic and buckling analyses of liquid storage tanks have been developed. Different aspects of nonlinear methodologies as well as the computer implementation aspects have been discussed. A finite element method computer code [47], which will be discussed in a separate report, has been developed on the basis of these investigations. A number of buckling and dynamic analyses of tanks have been made via this computer program, but, a full understanding of the behavior has not yet been achieved and requires further studies.

Finally, a word of caution is in order regarding the ability of the present capabilities to solve general problems of fluid and structural response. Although a degree of confidence has been established in certain realms of phenomena, it should be emphasized that the present developments are incapable of solving many problems of physical interest and application of these techniques in such circumstances is speculative at best. Areas which are particularly difficult and presently beyond the developed capabilities are (among others) high Reynolds number flows and complex shell buckling. Much research still needs to be done in these areas.

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References

- [1] A.A. Amsden and C.W. Hirt, Report LA-5100, Los Alamos Scientific Laboratory, Los Alamos, NM (March 1973).
- [2] H. Armen, *Comput. Structures* 10 (1979) 161-174.
- [3] J.H. Argyris and P.C. Dunne, Report 224, Institut für Statik und Dynamik der Luft und Raumfahrtkonstruktionen, University of Stuttgart, Germany (1977).
- [4] T. Belytschko and R. Mullen, Mesh partitions of explicit-implicit time integration, *Formulations and Computational Algorithms in Finite Element Analysis*, eds., K.J. Bathe, J.T. Oden and W. Wunderlich (M.I.T. Press, Cambridge, MA, 1977).
- [5] T. Belytschko and R. Mullen, *Internat. J. Numer. Meth. Engrg.* 12 (1978) 1575-1586.
- [6] T. Belytschko, H.J. Yen and R. Mullen, *Comput. Meths. Appl. Mech. Engrg.* 17/18 (1979) 259-275.
- [7] T. Belytschko and J.M. Kennedy, Computer models for subassembly simulation, preprint.
- [8] A. Brooks and T.J.R. Hughes, Steamline-upwind/Petrov-Galerkin methods for advection dominated flows, Third Internat. Conf. on Finite Element Methods in Fluid Flow, Banff, Canada, 1980.
- [9] I. Christie, D.F. Griffiths, A.R. Mitchell and O.C. Zienkiewicz, *Internat. J. Numer. Meth. Engrg.* 10 (1976) 1389-1396.
- [10] J. Donea, P. Fasoli-Stella and S. Giuliani, Lagrangian and Eulerian finite films techniques for transient fluid-structure interaction problems, *Trans. Fourth Internat. Conf. on Structural Mechanics in Reactor Technology*, San Francisco, August 1977.
- [11] J. Donea, P. Fasoli-Stella, S. Giuliani, J.P. Halleux and A.V. Jones, The computer code EURDYN-1M for transient dynamic fluid-structure interaction, EUR6751EN, Commission of the European Communities, Joint Research Center Ispra Establishment, Italy, 1980.
- [12] J. Donea, P. Fasoli-Stella, S. Giuliani, J.P. Halleux and A.V. Jones, paper No. B1/3, *Trans. SMIRT-5*, Berlin (West), Germany, August 13-17, 1979.
- [13] N. Edwards, A procedure for the dynamic analysis of thin-walled cylindrical liquid storage tanks, Ph.D. Thesis, University of Michigan, Ann Arbor, Michigan, 1969.
- [14] C.A. Felippa, K.C. Park and H.C. Yee, Synthesis of staggered solution procedures for coupled field analysis, 2nd Internat. Conf. Appl. Numerical Modeling, Polytechnical University of Madrid, Madrid, Sept. 11-15, 1978.
- [15] D.G. Goring, Report No. KH-R-38, California Institute of Technology, (Nov. 1978).
- [16] D.F. Griffiths, Research paper No. 317, University of Calgary, Calgary, Alberta, Canada (Sept. 1976).
- [17] J.O. Hallquist, Lawrence Livermore Laboratory, Report UCRL-52678, University of California, Livermore (March 3, 1979).
- [18] R.D. Hanson, Behavior of liquid storage tank, The Great Alaska Earthquake of 1964, *Engineering* (National Academy of Sciences, Washington, D.C. 1978) pp. 331-339.

- [19] M.A. Haroun, Dynamic analysis of liquid storage tanks, Ph.D. Thesis, California Institute of Technology, Pasadena, CA (1979).
- [20] J.C. Heinrich, P.S. Huyakorn, O.C. Zienkiewicz and A.R. Mitchell, *Internat. J. Numer. Meth. Engrg.* 11 (1977) 131–143.
- [21] L.M. Hoskins and L.S. Jacobsen, *Bull. SSA* 24 (1934).
- [22] G.W. Housner, *Bull. SSA* 47 (1957).
- [23] G.W. Housner, *Bull. SSA* 53 (1963).
- [24] G.W. Housner and M.A. Haroun, Seismic design of liquid storage tanks, preprints.
- [25] T.J.R. Hughes and A. Brooks, *AMD* 34 (ASME, New York, 1979) pp. 19–35.
- [26] T.J.R. Hughes, On consistently derived tangent stiffness matrices, to appear.
- [27] T.J.R. Hughes and K.S. Pister, *Comput. and Structures* 8 (1978) 391–397.
- [28] T.J.R. Hughes and J.M. Winget, Symmetric and nonsymmetric equation solvers, to appear.
- [29] T.J.R. Hughes and W.K. Liu, Nonlinear finite element analysis of shells: Part I—Three dimensional shells; Part II—Two-dimensional shells, *Comput. Meths. Appl. Mech. Engrg.*, to appear.
- [30] T.J.R. Hughes, W.K. Liu and A. Brooks, *J. Computational Phys.* 30 (1979) 1–60.
- [31] T.J.R. Hughes, W.K. Liu and T. Zimmermann, Lagrangian-Eulerian finite element formulation for incompressible viscous flows, U.S.-Japan Seminar on Interdisciplinary Finite Element Analysis, Cornell University, Ithaca, NY, August 7–11, 1979.
- [32] T.J.R. Hughes and J.M. Winget, Finite rotation effects in numerical integration of rate constitutive equations arising in large-deformation analysis, *Internat. J. Numer. Meth. Engrg.*, to appear.
- [33] T.J.R. Hughes, ed., *Finite Element Methods for Convection Dominated Flows*, *AMD* 34 (ASME, New York, Dec. 1979).
- [34] T.J.R. Hughes and W.K. Liu, *J. Appl. Mech.* 45 (1978) 371–374.
- [35] T.J.R. Hughes and W.K. Liu, *J. Appl. Mech.* 45 (1978) 375–378.
- [36] T.J.R. Hughes, K.S. Pister and R.L. Taylor, *Comput. Meths. Appl. Mech. Engrg.* 17/18 (1979) 159–182.
- [37] T.J.R. Hughes and R.A. Stephenson, Convergence of implicit-explicit algorithms in nonlinear transient analysis, *Internat. J. Engrg. Sci.*, to appear.
- [38] T.J.R. Hughes, T.K. Caughey and W.K. Liu, *J. Appl. Mech.* 45 (1978) 366–370.
- [39] T.J.R. Hughes, Implicit-explicit finite element techniques for symmetric and nonsymmetric systems, *Proc. Conf. on Nonlinear Problems in Mechanics*, University of Swansea, Swansea, U.K., Sept. 2–5, 1980.
- [40] L.S. Jacobsen and R.S. Ayre, *Bull. SSA* 41 (1951).
- [41] L.S. Jacobsen, *Bull. SSA* 39 (1949).
- [42] S.W. Key, Sandia Laboratories, Albuquerque, NM, SLA-74-0039 (April 1974).
- [43] R.D. Krieg, SAND 77-0943, Sandia Laboratories, Albuquerque, NM (1977).
- [44] R.D. Krieg and S.W. Key, Implementation of a time independent plasticity theory into structural computer programs, *Constitutive Equations in Viscoplasticity: Computational and Engineering Aspects*, *AMD* 20 (ASME, New York, 1976).
- [45] R.D. Krieg and D.B. Krieg, *ASME J. Pressure Vessel Technol.* 99 (1977) 510–515.
- [46] W.K. Liu, Development of finite element procedures for fluid-structure interaction, Ph.D. Thesis, California Institute of Technology, 1980.
- [47] W.K. Liu, FLUSTR, A fluid-structure interaction finite element analysis computer program, User's Manual, in preparation.
- [48] R.M. McMeeking and J.R. Rice, *Internat. J. Solids and Structures* 11 (1975) 601–616.
- [49] R.D. Mindlin, *J. Appl. Mech.* 18 (1951) 31–38.
- [50] N.M. Newmark, *J. Engrg. Mech. Div. ASCE* (1959) 67–94.
- [51] K.C. Park, C.A. Felippa and J.A. DeRuntz, Stabilization of staggered solution procedures for fluid-structure interaction analysis, *Computational Methods for Fluid-Structure Interaction Problems*, eds., T. Belytschko and T.L. Geero, *Applied Mechanics Symposia Series* (ASME, New York, 1977).
- [52] K.C. Park, Partitioned transient analysis procedure for coupled-field problems, presented at Second Conf. on Numerical Methods in Nonlinear Mechanics, TICOM, University of Texas, Austin, March, 1979.
- [53] K.C. Park and C.A. Felippa, *Comput. Meths. Appl. Mech. Engrg.* 17/18 (1979) 277–313.
- [54] K.C. Park and P.G. Underwood, *ASME Paper No. 79-PVP-120*, presented at the Pressure Vessels and Piping Conf., San Francisco, CA, June 25–29, 1979.
- [55] J.E. Rinne, Oil storage tanks, The Prince William Sound, Alaska, Earthquake of 1964, and Aftershocks, Vol. II, Part A, ESSA, U.S. Coast and Geodetic Survey (Washington, Government Printing Office, 1967) pp. 245–252.
- [56] S.H. Shaaban and W.A. Nash, Response of an empty cylindrical ground supported liquid storage tank to base excitation, *Depth. of Civil Engrg.*, University of Massachusetts (August 1975).
- [57] S.P. Timoshenko and J.M. Gere, *Theory of Elastic Stability* (McGraw-Hill, New York, 1961).
- [58] P.G. Underwood and K.C. Park, *ASME Paper No. 79-PVP-121*, presented at the Pressure Vessels and Piping Conference, San Francisco, CA, June 25–29, 1979.
- [59] A.S. Veletsos, Seismic effects in flexible liquid storage tank, *Proc. 5th World Conf. on Earthq. Engrg.* 1 (1974).
- [60] A.S. Veletsos and J.Y. Yang, Dynamics of fixed-base liquid-storage tanks, *Proc. U.S.-Japan Seminar on Earthquake Engrg. Research with Emphasis on Lifeline Systems*, Tokyo, Nov. 1976, pp. 317–341.
- [61] P.W. Werner, and K.J. Sundquist, *Trans. AM. Geophys. Union* 30 (1949).
- [62] H.M. Westergaard, *Trans. ASCE* 98 (1933).
- [63] J.P. Wright, *Comput. and Structures* 10 (1979) 235–238.
- [64] C.E. Wu, T. Mouzakis, W.A. Nash and J.M. Colonell, Natural frequencies of cylindrical liquid storage containers, *Dept. of Civil Engrg. University of Massachusetts* (June 1975).
- [65] J.Y. Yang, Dynamic behavior of fluid-tank systems, Ph.D. Thesis, Rice University, Houston, TX, 1976.